

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 1.9240876578635355, median 1.8027870469254028, std: 0.9526296634052777
Gyroscope error (imu0): mean 1.1242499701599782, median 0.744751932932354, std: 1.2179560643227705
Accelerometer error (imu0): mean 0.6698078832468308, median 0.4838910377276623, std: 0.6364782165250974

Residuals

Reprojection error (cam0) [px]: mean 1.9240876578635355, median 1.8027870469254028, std: 0.9526296634052777
Gyroscope error (imu0) [rad/s]: mean 0.09548361271918374, median 0.06325248567794702, std: 0.10344215987680049
Accelerometer error (imu0) [m/s^2]: mean 0.3672021652974679, median 0.26527880794761766, std: 0.34893017105109175

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[ [ 0.03072826 -0.99944486 0.01287432 -0.0047844 ]  
[ 0.99861196 0.03124881 0.04239877 0.00315065 ]  
[ -0.04277754 0.01155361 0.99901782 -0.01636564 ]  
[ 0. 0. 0. 1. ] ]
```

T_ic: (cam0 to imu0):

```
[ [ 0.03072826 0.99861196 -0.04277754 -0.00369935 ]  
[ -0.99944486 0.03124881 0.01155361 -0.00469111 ]  
[ 0.01287432 0.04239877 0.99901782 0.01627758 ]  
[ 0. 0. 0. 1. ] ]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.07979175866389675

Gravity vector in target coords: [m/s^2]

[-0.00444046 -9.80529393 0.15688913]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [567.2852783486742, 566.4961372581788]

Principal point: [652.7255421321942, 397.4199032103684]

Distortion model: equidistant

Distortion coefficients: [-0.05551145786489933, 0.029442090222052172, -0.028299846668714417,
0.00781929524076502]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.088 [m]

Spacing 0.026399999999999996 [m]

IMU configuration

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IMU0:

Model: scale-misalignment

Update rate: 100

Accelerometer:

Noise density: 0.05482201307

Noise density (discrete): 0.5482201307

Random walk: 0.04369065701339204

Gyroscope:

Noise density: 0.00849309453

Noise density (discrete): 0.08493094529999999

Random walk: 0.00011002624872473468

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

[0. 1. 0. 0.]

[0. 0. 1. 0.]

[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

Gyroscope:

M:

[[1.00796056 0. 0.]

[-0.04050531 0.93180704 0.]

[-0.04225121 -0.0439022 0.97431473]]

A [(rad/s)/(m/s²)]:

[[-0.00001972 0.00247025 -0.00185103]

[0.00002749 0.00100661 -0.00064421]

[0.0004573 -0.00525943 -0.00143199]]

C_gyro_i:

[[0.99925612 -0.01381317 0.03600556]

[0.01454801 0.99968955 -0.02022748]

[-0.03571498 0.02073624 0.99914686]]

Accelerometer:

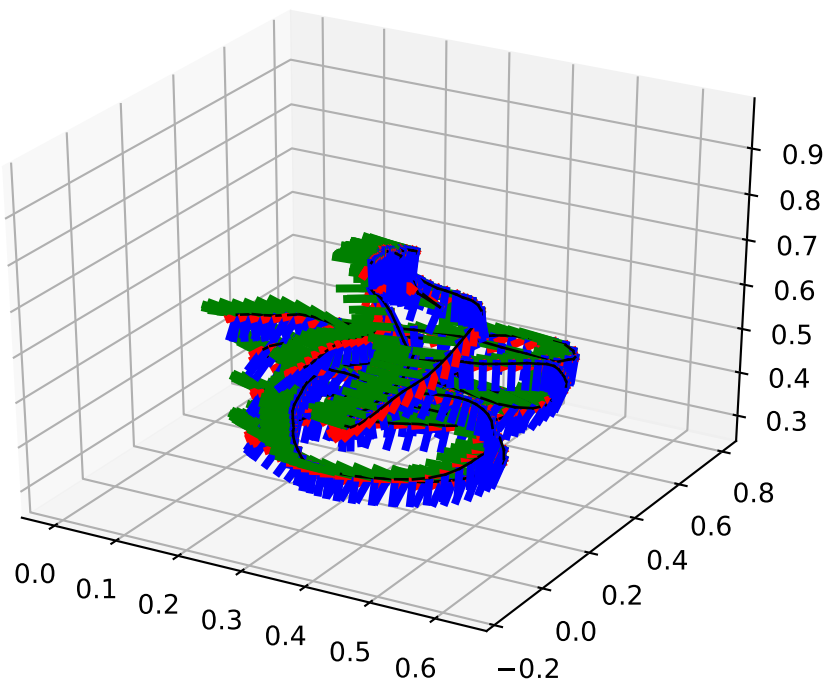
M:

[[0.98305839 0. 0.]

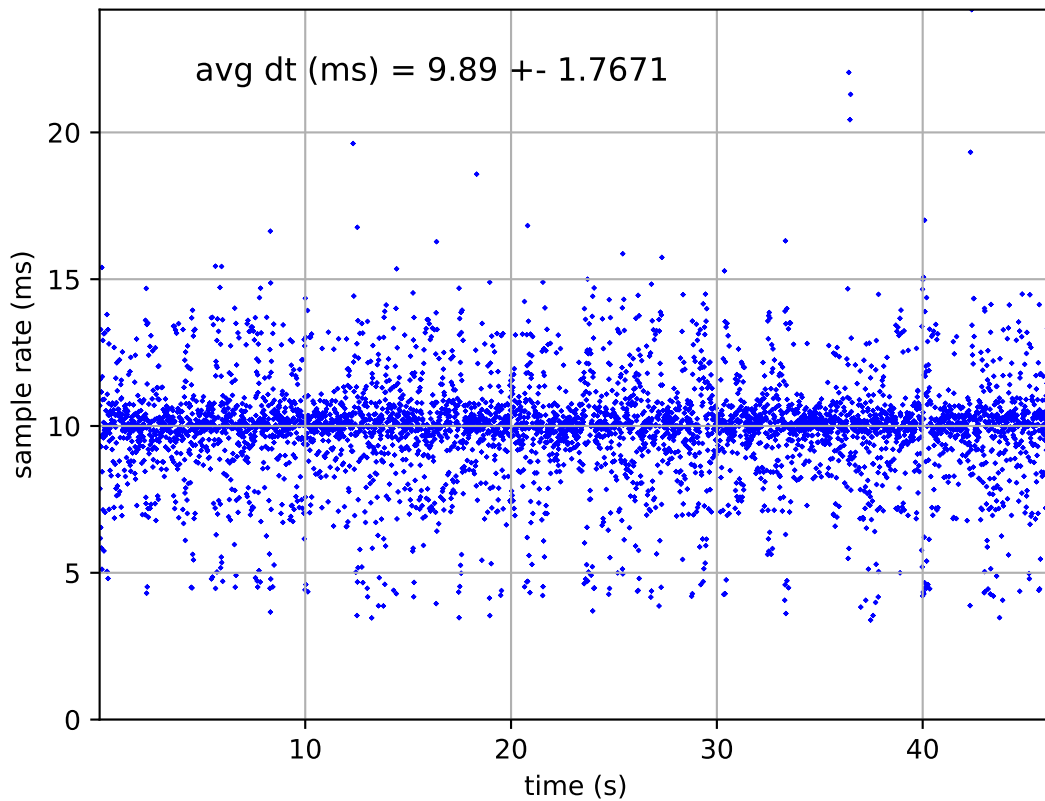
[-0.02385585 0.8930061 0.]

[0.01434692 0.02668485 -0.95160669]]

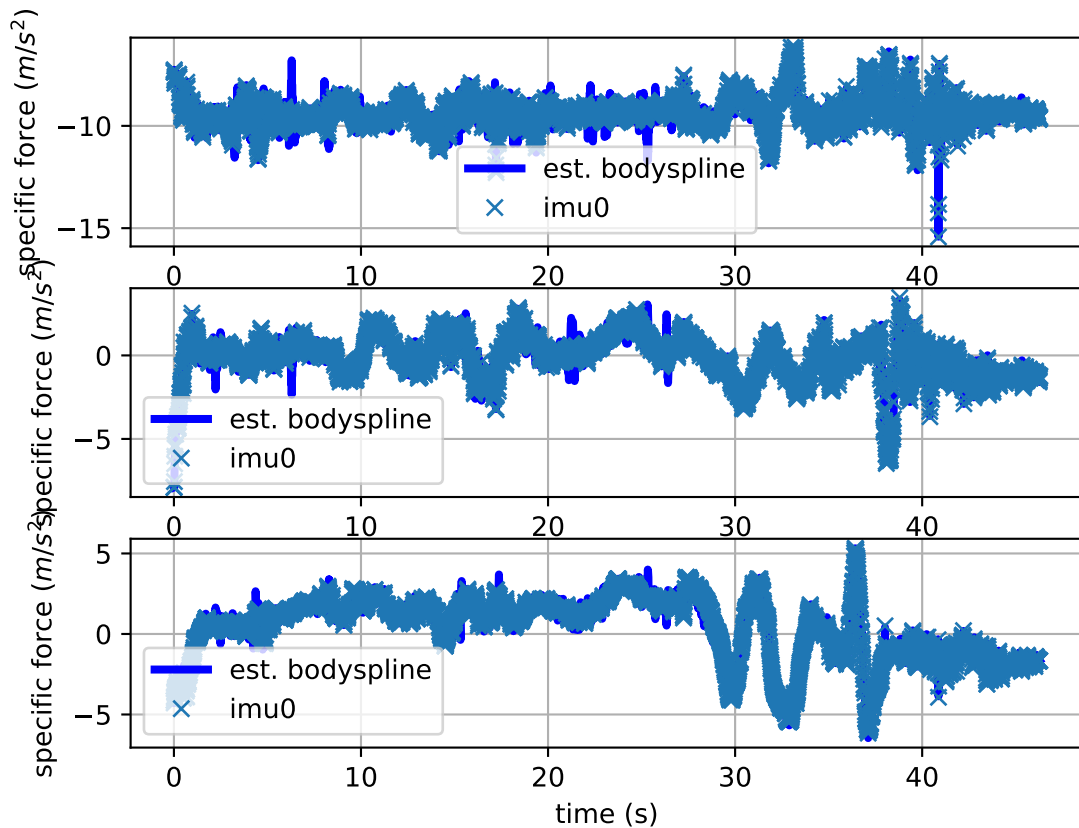
imu0: estimated poses



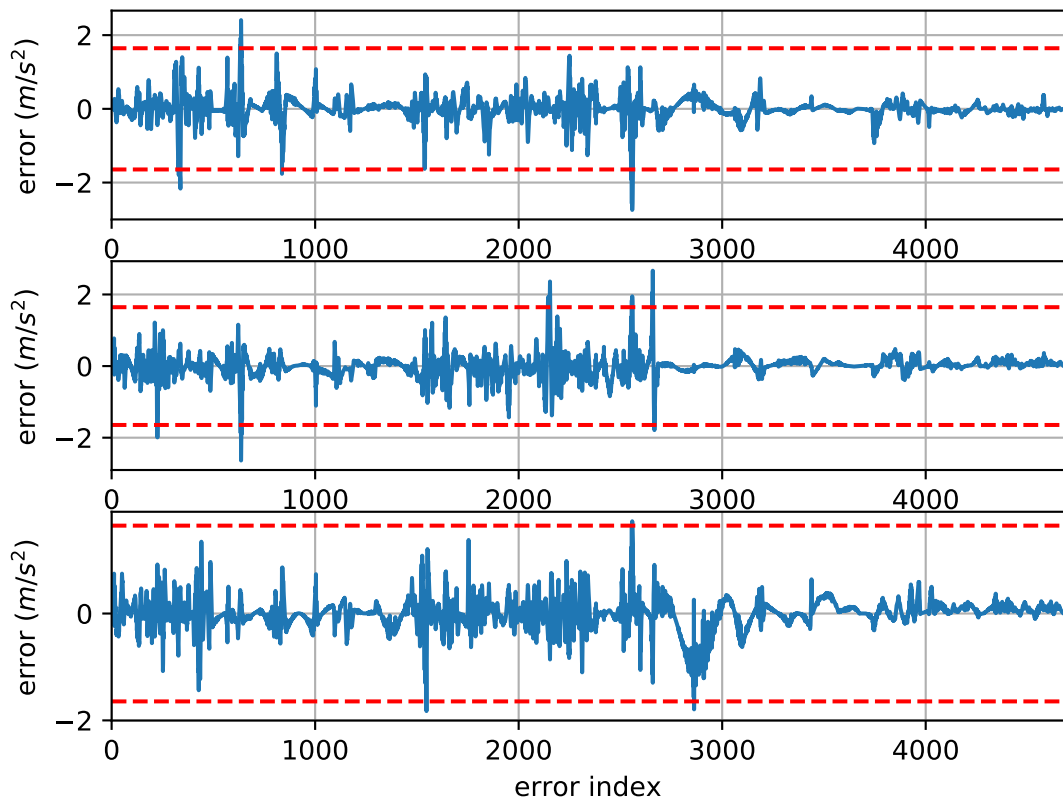
imu0: sample inertial rate



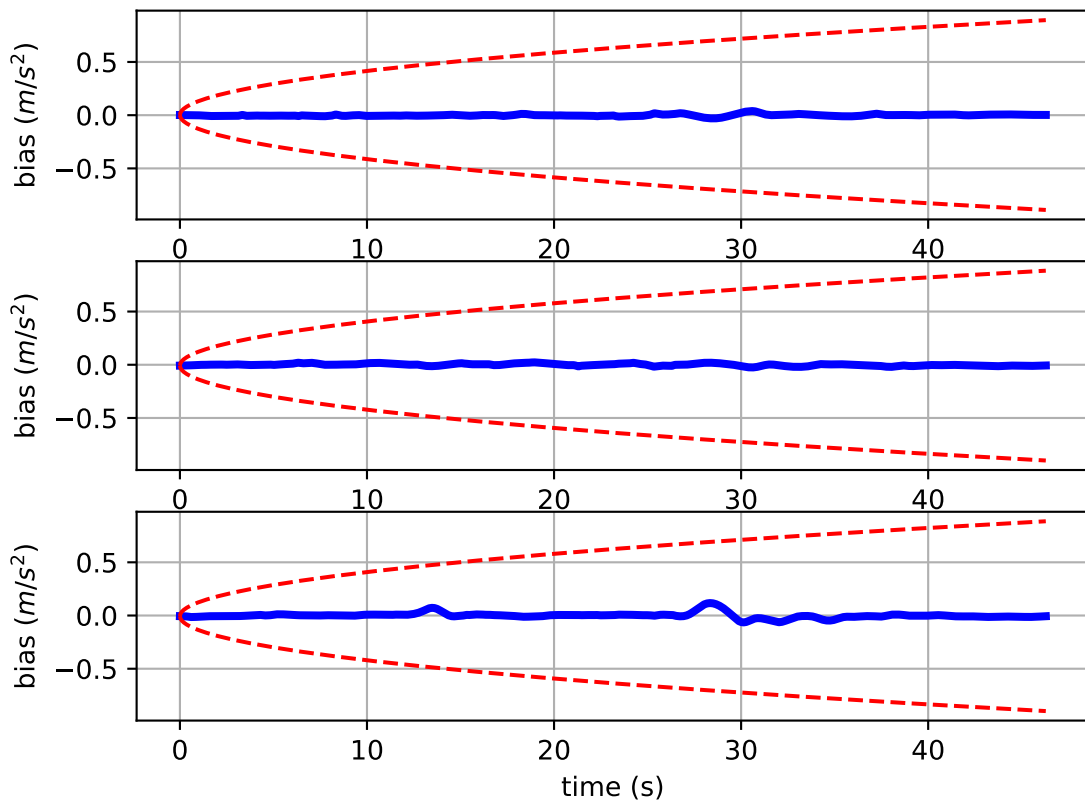
Comparison of predicted and measured specific force (imu0 frame)



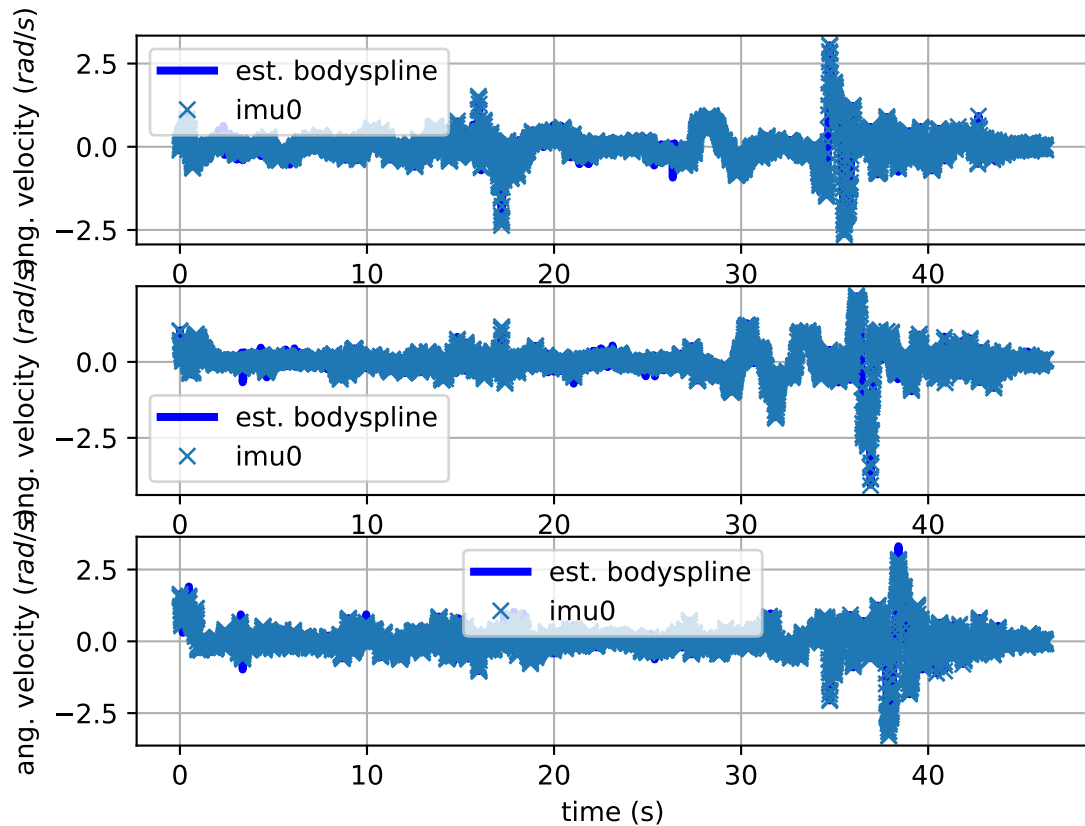
imu0: acceleration error



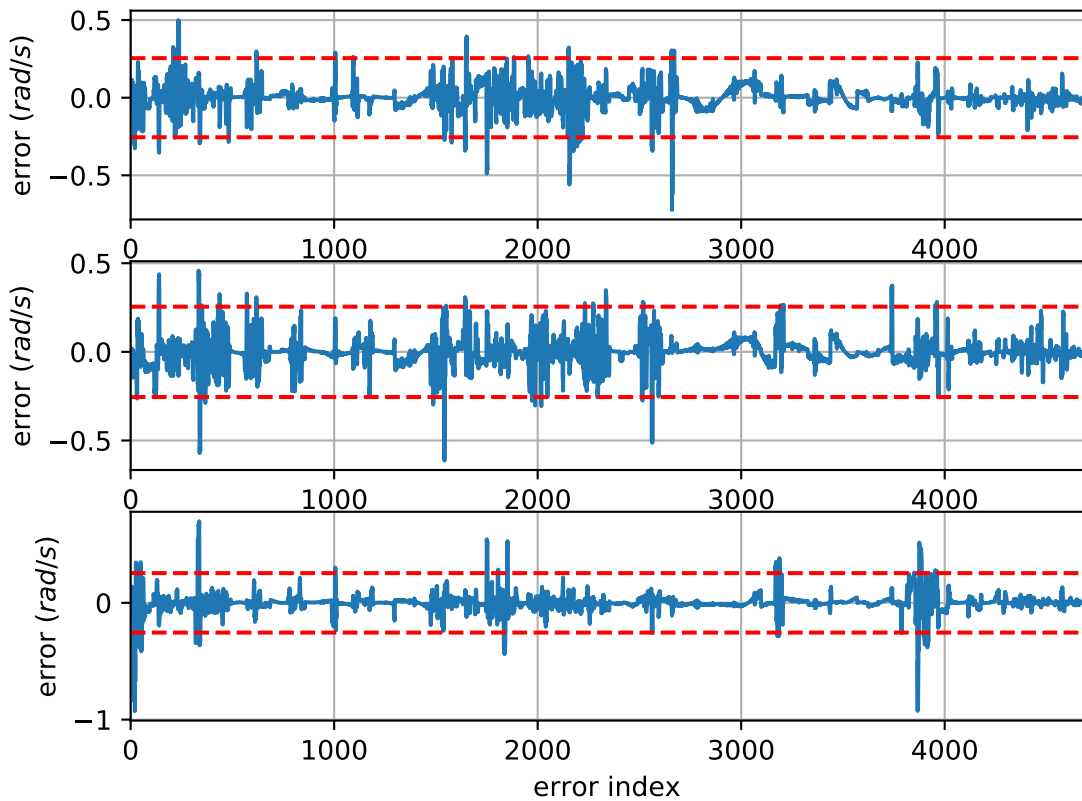
imu0: estimated accelerometer bias (imu frame)



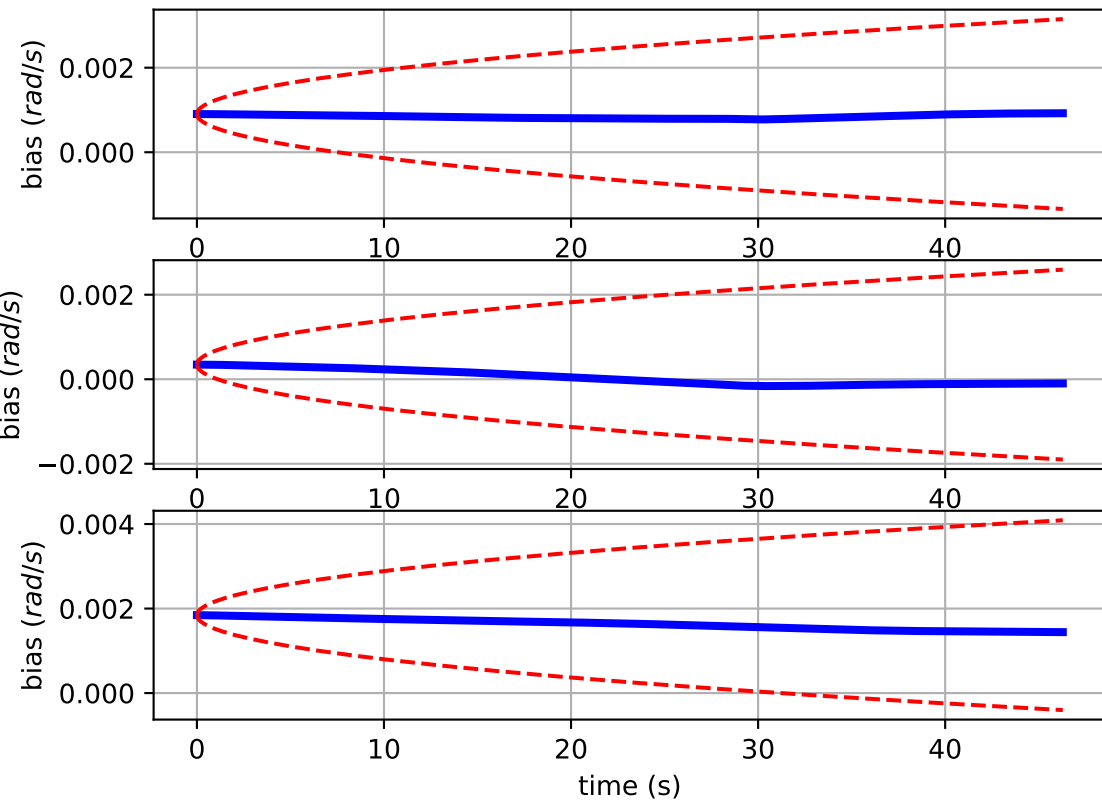
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

